

## Evidence Locker System — Execution & Demo Steps

Deployed URL: [Blockchain-Based-Evidence-Locker-System](#)

This guide contains the step-by-step instructions to run the project from scratch and demo its core features.

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### 1. Project Setup (First Time Only)

#### Install Backend Dependencies

1. Open a terminal.
2. Navigate to the backend folder:

```
cd backend
```

3. Run:

```
npm install
```

#### Install Frontend Dependencies

1. Open a new terminal.
2. Navigate to the frontend folder:

```
cd frontend
```

3. Run:

```
npm install
```

Ensure the .env file in the backend contains the correct:

- DATABASE\_URL
  - Cloudinary credentials
  - Email credentials
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### 2. Initialize Database & Users

In the backend terminal, run:

```
node config/seed.js
```

This initializes the database schema and creates the default users.

Expected output:

- Users created successfully
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### **3. Start the Application**

#### **Start Backend Server**

In the backend terminal, run:

```
npm run dev
```

Expected output:

- Server running on port 5000
- Connected to PostgreSQL

#### **Start Frontend Client**

In the frontend terminal, run:

```
npm run dev
```

Open the browser and go to:

```
http://localhost:3000/login
```

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### **4. Demo Workflow**

#### **A. Login & Register Evidence**

Login as Officer:

Email: officer@evidence.gov

Password: officer123

Steps:

1. Go to Register Crime.
  2. Fill out the crime details and submit.
  3. Open the created Crime Folder.
  4. Click + Register Evidence.
  5. Fill out the evidence details.
  6. Upload an evidence photo.
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#### **B. Custody Transfer**

1. Go to Custody Transfer.
2. Initiate a transfer:
  - Select the evidence
  - Transfer it to the Custodian (Jane Smith)

Logout and login as Custodian:

Email: `custodian@evidence.gov`

Password: `custodian123`

Steps:

1. Open Custody Transfer.
  2. Go to Scan QR Code.
  3. Scan the evidence QR code to accept custody.
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### **C. Forensic Report Upload**

Logout and login as Forensic Tech:

Email: `tech@evidence.gov`

Password: `tech123`

Steps:

1. Go to Forensic Reports.
  2. Select the evidence.
  3. Upload a PDF forensic report.
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## **5. Blockchain Tamper-Proofing Demo**

### **A. Simulate a Database Tamper Attack**

Keep the servers running.

Open a new terminal, navigate to the backend folder, and run:

```
node simulate_tamper.js
```

This script simulates a hacker directly modifying the PostgreSQL database.

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### **B. Run Integrity Audit**

Login as Admin:

Email: admin@evidence.gov

Password: admin123

Steps:

1. Go to Integrity Audit.
  2. Click Run Full System Audit.
  3. The system will detect the tampered block and mark the evidence as Corrupted.
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### **C. Restore Valid Copy**

1. On the Audit page, locate the corrupted evidence.
2. Click Restore Valid Copy.

The system will:

- Remove the tampered blocks
  - Preserve the valid evidence history
  - Keep uploaded photos and forensic reports
  - Generate a clean copy with a new Evidence ID and QR code
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### **6. Reset the System**

To clear all crimes, evidence, and logs while keeping user accounts:

```
node config/reset_db.js
```